CMSC 104 Section 2
Spring 2026
Sample Final Exam

Overview:

This exam is worth a total of 200 points. Consistent with the syllabus, it will count for 30% of the semester grade.

There are three sections. Section 1 consists of 15 True/False and Multiple Choice questions, each of which is worth 5 points, for a total of 75 points. There is no partial credit awarded for questions in Section 1. Section 2 consists of 10 short answer questions, each of which is worth 8 points, for a total of 80 points. Partial credit WILL be awarded for questions in this section, so please make an effort to answer each question. Section 3 consists of 3 coding questions. Each of these is worth 15 points, for a total of 45 points, and partial credit WILL be awarded for questions in this section. Again, please make an effort to answer each question.

Instructions:

This exam is open-book, open-notes. You may use your laptop to refer to previous lecture notes, quizzes, coding assignments, etc. You may use an IDE to test code, to ensure that it works the way you intend. However, ALL answers must be written on the test paper in order to be graded.

You MAY NOT use any AI tool, such as ChatGPT, Google Gemini, Microsoft CoPilot, etc. to answer any part of any question.

You MAY NOT interact with any other student, whether inside or outside of the classroom, during this exam. That includes texting or messaging other students - that is strictly prohibited.

Section 1: True/False and Multiple Choice

1. Which of the following is a valid Python variable?
    a. `2K00L4SK00L`
    b. `__turn_left`
    c. `Left-turn`
    d. None of the above is a valid Python variable

2. What is the value of the following Python expression?
    ```
    y = (5**2)and not (3>(5//2))
    ```
    a. There is no value; this is not a valid Python expression
    b. 3
    c. False
    d. True

3. What is printed out as a result of this loop of Python code?
    ```
    sum = 1
    for i in range(1,10,3):
       sum -= i
    print (sum)
    ```
    a. 11
    b. 10
    c. -11
    d. None of the above is correct

4. True or False: the “while” loop is the most general type of loop in Python. That means that any problem that can be solved with a loop can be solved with a while loop. The same can not be said of “for” loops.
    a. True
    b. False

5. Which of the following represents “polymorphism” in Object oriented programming?
    a. When a class and its subclass have methods with the same name, that do different things for a superclass member and a subclass member
    b. When a class method can return any random result for any given object
    c. When a subclass and its superclass have the identical method that returns identical results
    d. There’s no such word as “polymorphism,” I made that up to trick you

6. What is returned by a function if there is no return statement in the function's code?
    a. The program will crash
    b. Nothing
    c. The special value None, of type NoneType
    d. None of the above is correct.

7. What is the difference between calling a function with immutable parameters such as ints or strings, and calling a function with mutable parameters such as lists or dictionaries?
    a. There is no difference
    b. Calling a function using mutable parameters is forbidden in Python
    c. Calling using immutable types uses call-by-value, mean changes made in the function ARE NOT reflected in the main program variables. Calling using mutable types uses call-by-reference, meaning when you change the parameter's value in the function, that change IS reflected in the main program variable, whether you wanted it to be or not.
    d. None of the above is correct.

8. True or False: strings are mutable types. To change the string s = "UMBC" to s="UMB" you would just type
    ```
    s.remove("C")
    ```
    a. True
    b. False

9. What is the difference between using append() to add an item to a list and using insert() to add the item?
    a. append() puts the item at the end of the list, while insert() puts it where you tell it to
    b. append() puts the item at the beginning of the list, while insert() puts it where you tell it to
    c. insert() puts the item at the end of the list, while append() puts the item where you tell it to
    d. None of the above is correct

10. True or False: a Python dictionary is an ordered data structure. That is, it matters in what order you enter the key/value pairs.
    a. True
    b. False

11. What is the best way to develop software?
    a. Write a little, test a little
    b. Write the whole thing at once and hope it works
    c. Use an AI app like ChatGPT or Claude
    d. Come on, you know this by now

12. True or False: you can have an if-elif code segment without an else clause. That is, the following is valid Python code:

```
if age > 21:
   print("Hooray, fully adult")
elif age > 18:
   print("Partway there; not quite")
```

   a. True
   b. False

13. What happens if you try to open a file for reading ("r" mode) in Python and the file doesn't exist?
   a. Python automatically creates a new empty file for you
   b. Your program crashes
   c. Python opens the first file in whatever directory/folder your program is executing
   d. None of the above is correct.

14. True or False: when you open a file for writing ("w" mode) in Python, any existing data in the file is deleted and you start with an empty file. So accidentally opening the file you want to read in "w" mode is a bad thing.
   a. True
   b. False

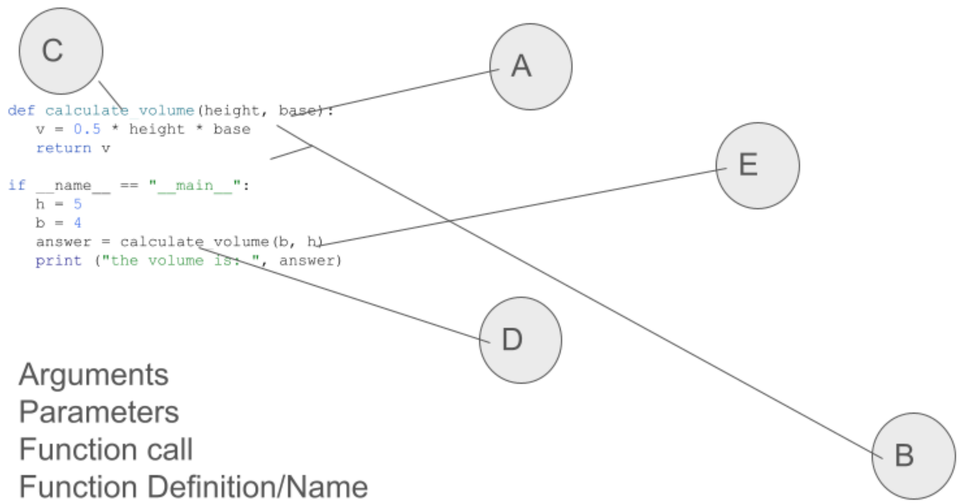
15. Suppose you have the following string in Python. How do you print out just the digit 1?

```
motto = "CMSC 104 is the best elective"
```

   a. print(motto[4])
   b. print(motto[5])
   c. print(motto[3])
   d. None of the above would work

Section 2: Short Answer

16. Identify each of the parts of a function below



Arguments
Parameters
Function call
Function Definition/Name
Function Body

17. Suppose that you have the following list of states:

```
states = ["Alabama", "Alaska", "Arizona", "Arkansas", "California", "Colorado",
"Connecticut"]
```

Write a line of Python code that will print out ONLY the first three states. (You should be able to do this with just a 'print' statement.)

18. There are two sections of CMSC 104 planned for the Fall semester. This dictionary represents the planned sections with the planned instructor for each:

```
cmsc104 = {
   1:"Zak",
   2:"Arsenault"
}
```

Due to increasing enrollment, a third section will be needed, with Grace Hopper hired as the instructor. Write a line of Python that adds this new key/value pair to the dictionary.

19.Suppose you have the following Python program:

```
def calculate_volume(height, base):
   v = 0.5 * height * base

if __name__ == "__main__":
   h = 5
   b = 4
   answer = calculate_volume(b, h)
   print ("the volume is: ", answer)
```

What is printed as the result of this program? Why?

20. What two types that we have learned about this semester are mutable types in Python?

21. Write Python code using two nested loops that creates a two-dimensional list with five rows and five columns, containing the digits from 0 through 24, inclusive. That is, the first row should have 0, 1, 2, 3 and 4; the next row 5, 6, 7, 8, and 9 and so on. You MUST use two nested loops to do this. You can use either for or while loops; your choice.

22. Suppose that we have a list,

```
l = [1,2,3,4]
```

What is the difference between

```
l.pop(3)
```

and

```
l.remove(3).
```

You can show the contents of l after each operation or just explain. .

23. Explain why, for any two integers x and y, x//y has to be less than or equal to x/y.

24. What is a sentinel while loop?

25. What happens if you open a file for append (‘a’) and the file doesn’t exist?

Section 3: Coding

26. Write a Python program that calculates the hypotenuse of a right triangle, using the formula $a^2 + b^2 = c^2$. C is the length of the hypotenuse; a and b are the lengths of the other two sides.

Your program must ask the user to enter the lengths a and b, which are positive floating point numbers. You do not need to do error checking; you can assume the user will enter positive floating point numbers.

Once you have these numbers, square each of them and add them together to get c^2. Then take the square root of c^2, and print out the result, labeled as being the hypotenuse of the triangle.

27. Write a Python function that calculates the volume of a sphere, using the formula volume = (4/3) * pi * radius.

Your main program must ask the user for the radius of the sphere, which will be a positive floating point number. No error checking is needed; presume the user will enter a positive floating point number.

Then you must call a function, passing the radius as an argument. Your function must take this value as its one parameter, and calculate the volume using the formula above. Then your function must return the calculated volume.

Your main program must print out the computed volume, labeled as such so the user understands.

Note: you can either import the math library and use math.pi in the formula, or you can explicitly set the value of pi in your program to 3.14159. Either is acceptable.

28. I have a file, medal_table.txt, that contains the number of medals won by each country at the latest World Aquatics (i.e., swimming and diving) Championships. The first line of the file is a list of column headings and contains no useful data. After that, each line contains the name of the country, the number of gold medals won, the number of silver medals won, and the number of bronze medals won. Each value is separated from each other by one tab ('\t') character.

Write a Python program that reads in this data file. Throw the first line away. For each remaining line, split the line into the four relevant values; convert the numbers of gold, silver and bronze medals into integers, and add the numbers together to the get the total number of medals the country won.

Then print out the name of the country and the total number of medals won.